Ho Thinh Phat

hothinhphat06@gmail.com • 0915623625 • https://github.com/miyxotkem

Education

University of Information Technology - Vietnam National University HCMC — **Ho Chi Minh City**
Bachelor of Software Engineering — 2024 - present
GPA (as of 4th semester): 8.77/10

Publications

LogTransAD: Adaptive Transformer-Based Log Anomaly Detection for Reliable Cloud and Networked Systems — **UIT**
Author — 2026
Technologies: Python, PyTorch, Transformers, LSTM, Drain, DBSCAN
Key Responsibilities & Achievements:

- Authored a scientific research paper on AIOps for incident detection, proposing a novel framework (LogTransAD) for cloud and microservice systems, accepted at the 20th 10th IEEE Symposium on Wireless Technology & Applications (ISWTA 2026).
- Engineered a sequence modeling pipeline that parses raw logs using Drain, replacing traditional sequential LSTM models with a multi-head self-attention Transformer (TLS) to better capture long-range event dependencies.
- Designed and evaluated an Adaptive Top-K (ATK) entropy-driven thresholding mechanism that dynamically adjusts detection sensitivity during non-stationary workload transitions, such as software deployments.
- Benchmarked the framework against DeepLog-LSTM baselines using a 500,000-line HDFS log subset and a synthetic Deployment Event Simulation (DES) stress test, achieving a 0.804 F1-score and 0.995 recall.

Projects

Multi-VLM Ensemble Pipeline for UAV Domain-Shift Auto-Labeling - Test-Time Adaptation Research — **UIT**
AI Researcher — 2026 - present
Technologies: Python, PyTorch, Hugging Face Transformers, BitsAndBytes (4-bit quantization), Pandas, Slurm/HPC
Description: Designed and built a hybrid annotation pipeline that uses an ensemble of open-source vision-language models to automatically classify domain-shift attributes (viewpoint, illumination, motion blur, background complexity, occlusion) in aerial drone imagery, replacing costly manual labeling for large-scale domain-shift analysis on DOTA and VisDrone.
Key Responsibilities & Achievements:

- Orchestrated 3 large VLM workers (Qwen3.8-27B, Gemma-4-31B, InternVL3.5-30B) as independent "expert annotators," each reasoning over a defined taxonomy and diagnostic cues before predicting labels.
- Designed a 3-tier ensemble router: majority voting for high-agreement/simple attributes, calibrated weighted-sum scoring for harder attributes, and escalation to a "judge" VLM (GLM-4.1V-9B-Thinking) when models disagreed or confidence was low — a Mixture-of-Agents-style consensus mechanism.
- Solved GPU memory constraints by applying 4-bit (NF4) quantization via BitsAndBytes to run three ~27–31B

parameter VLMs sequentially on a single shared GPU, and patched several transformers library bugs (tied-weights, OOM during warmup) to stabilize inference.

- Calibrated ensemble voting weights via per-model, per-attribute accuracy against a 200-image ground-truth set, and deployed all inference jobs on a Slurm-based HPC cluster with custom environment/CUDA diagnostics tooling.

SmartEdu Microservices - SE361 **UIT**
Backend Developer 2026
Technologies: C#, ASP.NET Core Web API, PostgreSQL, Redis, RabbitMQ, YARP API Gateway, Docker, Git.
Description: A backend system for an E-learning platform, rebuilt as 7 independently deployable microservices behind a unified API Gateway for scalability and fault tolerance.
Key Responsibilities & Achievements:

- Separated core business logic into independent, easily maintainable microservices (e.g., Course, Identity, and Payment services).
- Implemented a unified YARP API Gateway as the single entry point for all microservices, handling request routing, rate limiting, and WebSocket proxying.
- Configured RabbitMQ to handle asynchronous communication and event-driven data flow between different microservices.
- Integrated Redis for distributed caching to minimize database query latency and improve API response times.
- Containerized the application environment using Docker, ensuring consistent deployment and isolated execution of services and databases.

Leadership & Activities

myInsight MDS Datathon Challenge **FTU**
Team Leader 2026

- Led a student team (UIT-TraDaoCamSa) through a data-focused competition, managing schedules, task delegation, and problem-solving strategies.

University Student Ambassador **UIT**
Outreach Campaign Leader 2025-2026

- Led a team of student ambassadors on outreach visits to local high schools, coordinating event logistics and engagement activities with prospective students.

Technical Skills

Languages: IELTS (2023): 7.0, C#, Java, Python, C/C++.
Frameworks & Web: .NET, ASP.NET Core, WPF, WinForms.
Databases: SQL Server, PostgreSQL, Redis, Firebase (Cloud Firestore, Realtime DB).
Tools & Architecture: Git/GitHub, Docker, RabbitMQ, Postman, Microservices.
AI/ML: PyTorch, Hugging Face Transformers, BitsAndBytes, VLMs (Qwen, Gemma, InternVL), Slurm/HPC.